

# HKCEE 1980 Mathematics II

Label 1  $2ab = a^2 - b^2 \Rightarrow$

- A.  $(a-b)^2$   
 B.  $(-a-b)^2$   
 C.  $(-a+b)^2$   
 D.  $-(a+b)^2$   
 E.  $-(a-b)^2$

Label 2  $125^a \cdot 5^b =$

- A.  $625^{a+b}$   
 B.  $625^{ab}$   
 C.  $125^{a+3b}$   
 D.  $5^{a+3b}$   
 E.  $5^{3a+b}$

Label 3 If  $4p = 9q$ , then  $\frac{4p^2}{9q^2} =$

- A. 1  
 B.  $\frac{4}{9}$   
 C.  $\frac{9}{4}$   
 D.  $(\frac{4}{9})^2$   
 E.  $(\frac{9}{4})^2$

Label 4 If  $n = 10^a$ , then  $\log_{10} n =$

- A.  $10^a$   
 B.  $10^n$   
 C.  $n^a$   
 D.  $a^n$   
 E.  $a$

Label 5  $\frac{x^{-2} - y^{-2}}{x^{-1} - y^{-1}} =$

- A.  $x^{-1} + y^{-1}$   
 B.  $x^{-1} - y^{-1}$   
 C.  $x^{-3} - y^{-3}$   
 D. 1  
 E.  $\frac{1}{x+y}$

Label 6 If  $\frac{1}{x} = a + b$  and  $\frac{1}{y} = a - b$ , then  $x + y =$

- A. 2  
 B.  $\frac{a}{a^2 - b^2}$   
 C.  $\frac{a}{a^2 - b^2}$   
 D.  $\frac{2a}{a^2 - b^2}$   
 E.  $\frac{-2a}{a^2 - b^2}$

Label 7 If  $x = \frac{y + (n-1)z}{n+1}$ , then  $n =$

- A.  $\frac{x - y + z}{z}$   
 B.  $\frac{x + y - z}{z}$   
 C.  $\frac{y - x - z}{x + z}$   
 D.  $\frac{y - x - z}{x - z}$   
 E.  $\frac{y + x - z}{x - z}$

Label 8  $\frac{5^{n+2} - 35(5^{n-1})}{18(5^{n+1})} =$

- A.  $\frac{1}{18}$   
 B.  $\frac{1}{15}$   
 C.  $\frac{1}{5}$   
 D.  $5$   
 E.  $5^n$

Label 9 Solve the inequality

2.  $(4x + 3)(x - 4) > 0$

- A.  $x \geq 4$   
 B.  $4 > x > -\frac{3}{4}$   
 C.  $-\frac{3}{4} \geq x$   
 D.  $-\frac{3}{4} \geq x$  or  $x \geq 4$   
 E.  $x > -\frac{3}{4}$

Label 10 When the hour hand has turned through an angle  $x^\circ$ , what is the angle through which the minute hand has turned?

- A.  $6x^\circ$   
 B.  $12x^\circ$   
 C.  $60x^\circ$   
 D.  $360x^\circ$   
 E.  $3\,600x^\circ$

Label 11 The first term of an arithmetic progression is 6 and its tenth term is three times its second term. The common difference is

- A. 8  
 B. 4  
 C. 3  
 D. 2  
 E. 1

Label 12 A man sold a car for \$35 000 at a loss of 30% on the cost price. What would have been the loss or gain percent if he had sold it for \$50 500?

- A. A gain of 10%  
 B. A gain of 1%  
 C. No gain nor loss  
 D. A loss of 10%  
 E. A loss of 1%

Figure 12 Density 5045

Label 13 If the length of a rectangle is increased by 10% and the width decreased by 10%, which of the following is true?

- A. Its area remains the same  
 B. Its area is decreased by 1%  
 C. Its area is increased by 1%  
 D. Its area is decreased by 10%  
 E. Its area is increased by 10%

Label 14 The length of a side of a rhombus is 10 cm. If its shorter diagonal is of length 12 cm, what is the area of the rhombus in  $\text{cm}^2$ ?

- A. 60  
 B. 96  
 C. 100  
 D. 120  
 E. 192

Label 15 If the bearing of B from A is  $S30^\circ W$ , then the bearing of A from B is

- A.  $N30^\circ E$   
 B.  $N60^\circ W$   
 C.  $N60^\circ E$   
 D.  $S30^\circ W$   
 E.  $S30^\circ E$

Label 16 Figure 16 Density 1598

$$\frac{1}{\sin \theta} - 1 \quad \frac{1}{\sin \theta} + 1$$

- A.  $2 \tan \theta$   
 B.  $2 \tan^2 \theta$   
 C.  $\frac{2}{\tan^2 \theta}$   
 D.  $\frac{2 \sin \theta}{\cos^2 \theta}$   
 E.  $\frac{2 \sin^2 \theta}{\cos \theta}$

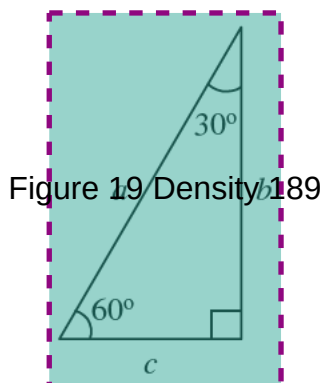
Label 17 If  $\cos \theta = x$  and  $0^\circ < \theta < 90^\circ$ , then  $\tan \theta$

- A.  $\frac{1}{\sqrt{1-x^2}}$
- B.  $\sqrt{1-x^2}$
- C.  $\frac{\sqrt{1-x^2}}{x}$
- D.  $\frac{x}{\sqrt{1-x^2}}$
- E.  $\pm \frac{x}{\sqrt{1-x^2}}$

Label 18 If  $0^\circ \leq \theta < 360^\circ$ , which of the following equations has exactly one root?

- A.  $\sin \theta = -1$
- B.  $\sin \theta = -\frac{1}{2}$
- C.  $\sin \theta = 0$
- D.  $\sin \theta = \frac{1}{2}$
- E.  $\sin \theta = 2$

Label 19



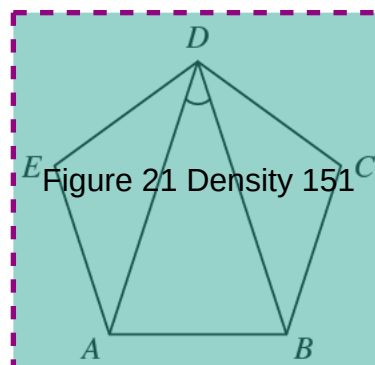
In the figure,  $a : b : c =$

- A.  $3 : 2 : 1$
- B.  $9 : 4 : 1$
- C.  $2 : \sqrt{3} : 1$
- D.  $\sqrt{3} : \sqrt{2} : 1$
- E.  $\sqrt{3} : 2 : 1$

Label 20 What is the area, in  $\text{cm}^2$ , of an equilateral triangle of side  $x$  cm?

- A.  $\frac{\sqrt{3}}{4}x^2$
- B.  $\frac{\sqrt{3}}{2}x^2$
- C.  $\frac{1}{4}x^2$
- D.  $\frac{1}{2}x^2$
- E.  $\sqrt{3}x^2$

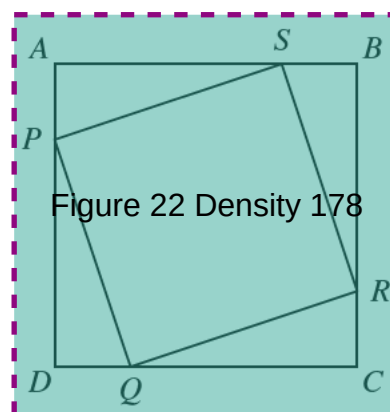
Label 21



In the figure,  $ABCDE$  is a regular pentagon.  $\angle ADB =$

- A.  $35^\circ$
- B.  $36^\circ$
- C.  $40^\circ$
- D.  $54^\circ$
- E.  $72^\circ$

Label 22

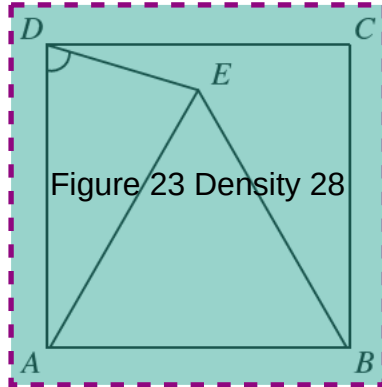


In the figure,  $ABCD$  is a square with  $AB = 5$ .  $AP = BQ = CR = DS = 1$ . What is the area of  $PQRS$ ?

- A. 9

- B. 15
- C. 16
- D. 17
- E. 18

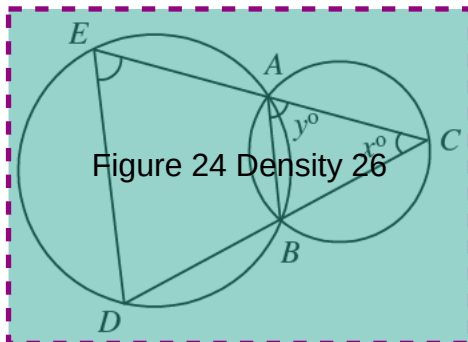
Label 23



In the figure,  $ABCD$  is a square and  $ABE$  is an equilateral triangle.  
 $\angle ADE = ?$

- A.  $72^\circ$
- B.  $74^\circ$
- C.  $76^\circ$
- D.  $78^\circ$
- E. None of the above

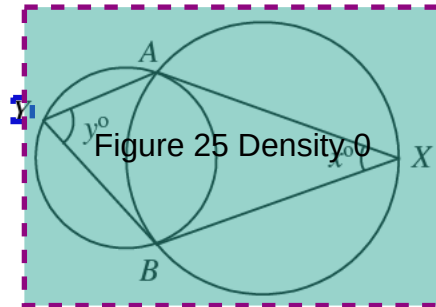
Label 24



In the figure, the two circles intersect at  $A$  and  $B$ .  $CAE$  and  $CBD$  are straight lines.  $\angle CED =$

- A.  $y^\circ$
- B.  $180^\circ - y^\circ$
- C.  $180^\circ - x^\circ - y^\circ$
- D.  $180^\circ - x^\circ + y^\circ$
- E.  $360^\circ - x^\circ - y^\circ$

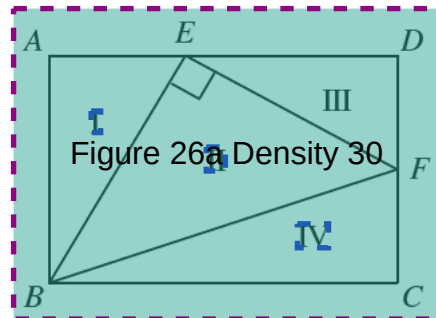
Label 25



In the figure, circle  $AXB$  passes through the centre of circle  $AYB$ .  $y =$

- A.  $2x$
- B.  $180 - 2x$
- C.  $180 - x$
- D.  $\frac{1}{2}(90 - x)$
- E.  $\frac{1}{2}(180 - x)$

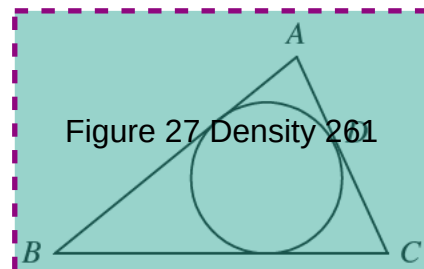
Label 26



In the figure,  $ABCD$  is a rectangle.  $\angle BEF = 90^\circ$ . Which two of the triangles I, II, III and IV must be similar?

- A. I and II
- B. I and III
- C. II and III
- D. II and IV
- E. III and IV

Label 27



In the figure, the inscribed circle of  $\triangle ABC$  touches  $AC$  at  $D$ . If  $AB = 7$ ,  $AC = 5$  and  $AD = 2$ , then  $BC =$

- A. 9.5
- B. 9
- C. 8.5
- D. 8
- E. 7.5

A certain sum of money is just sufficient to pay the wages of one man for  $m$  days or the wages of one boy for  $n$  days. For how many days will this sum be just sufficient to pay the wages of one man and one boy together?

- A.  $m+n$
- B.  $\frac{m+n}{2}$
- C.  $\frac{1}{m} + \frac{1}{n}$
- D.  $\frac{m+n}{mn}$
- E.  $\frac{mn}{m+n}$

If the value of  $y^2 + 3y + 7$  is 2, what is the value of  $2y^2 + 6y - 3$ ?

- A. -13
- B. -7
- C. 7
- D. 13
- E. It cannot be found from the information given

A, B, C are three spheres. If

$$\frac{\text{Surface area of } A}{\text{Surface area of } B} = 4 \text{ and } \frac{\text{Volume of } B}{\text{Volume of } C} = 2, \text{ then } \frac{\text{Volume of } A}{\text{Volume of } C} =$$

- A. 16
- B. 8
- C. 2

- D.  $\frac{1}{8}$
- E.  $\frac{1}{16}$

The  $2n^{\text{th}}$  term of the geometric progression,  $8, -4, 2, -1, \dots$ , is

- A.  $\frac{1}{2^{2n+2}}$
- B.  $\frac{-1}{2^{2n+2}}$
- C.  $\frac{1}{2^{2n-3}}$
- D.  $\frac{1}{2^{2n-4}}$
- E.  $\frac{-1}{2^{2n-4}}$

Figure 32

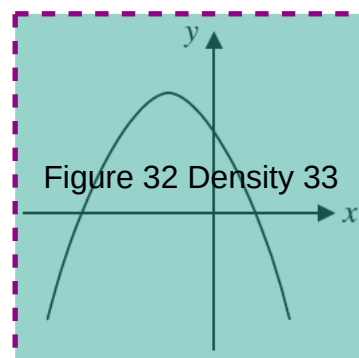


Figure 32 Density 33

The figure above shows the graph of  $y = ax^2 + bx + c$ . Determine whether  $a$  and  $c$  are positive or negative.

- A.  $a > 0$  and  $c > 0$
- B.  $a < 0$  and  $c < 0$
- C.  $a > 0$  and  $c < 0$
- D.  $a < 0$  and  $c > 0$
- E. It cannot be determined from the given data

$\$P$  amounts to  $\$Q$  in  $n$  years at simple interest. The rate per annum is

- A.  $\frac{100n(Q-P)}{P} \%$
- B.  $\frac{100(Q-P)}{n} \%$

Figure 33 Density 2459

- C.  $\frac{100(Q-P)}{nP} \%$   
 D.  $\frac{100(Q-P)}{nQ} \%$   
 E.  $100[(\frac{Q}{P})^{\frac{1}{n}} - 1] \%$

- A.  $k = -6$  only  
 B.  $k = 6$  only  
 C.  $-6 \leq k \leq 6$   
 D.  $k = 6$  or  $-6$  only  
 E.  $k \leq -6$  or  $k \geq 6$

Label 34  
 34 If  $0 < x < 1$ , which of  $x$ ,  $x^2$ ,  $\frac{1}{x}$ ,  $\sqrt{x}$  is the smallest? Which is the largest?

- A.  $\sqrt{x}$  is smallest,  $x^2$  is largest  
 B.  $\frac{1}{x}$  is smallest,  $x^2$  is largest  
 C.  $x$  is smallest,  $\frac{1}{x}$  is largest  
 D.  $x^2$  is smallest,  $\frac{1}{x}$  is largest  
 E.  $x^2$  is smallest,  $\sqrt{x}$  is largest

Label 38  
 38 If  $x$  and  $y$  are real numbers, what is the minimum value of the expression  $(x+y)^2 - 1$ ?

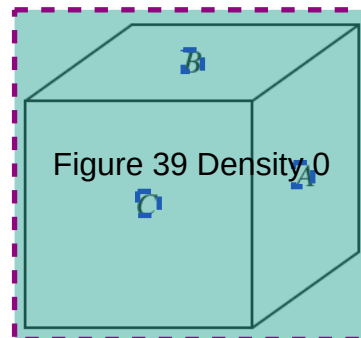
- A.  $-5$   
 B.  $-1$   
 C.  $0$   
 D.  $5$   
 E. It cannot be determined

Label 35  
 35 The Highest Common Factor of two unequal Positive integers  $a$  and  $b$  is 8. Which of the following must be true?

- I. The difference between  $a$  and  $b$  is divisible by 8  
 II.  $(a+b)$  is divisible by 16  
 III.  $ab$  is divisible by 64

- A. III only  
 B. I and II only  
 C. I and III only  
 D. II and III only  
 E. I, II and III only

Label 39  
 39



In the figure, the areas of the surfaces  $A$ ,  $B$ ,  $C$  of the cuboid are  $10 \text{ cm}^2$ ,  $14 \text{ cm}^2$  and  $35 \text{ cm}^2$  respectively. What is the volume of the cuboid?

- A.  $49 \text{ cm}^3$   
 B.  $70 \text{ cm}^3$   
 C.  $140 \text{ cm}^3$   
 D.  $350 \text{ cm}^3$   
 E.  $4900 \text{ cm}^3$

Label 36  
 36  $x$ ,  $y$  and  $z$  are three consecutive positive integers. Which of the following is true?

- A.  $x+y+z$  must be odd  
 B.  $x+y+z$  must be even  
 C.  $xyz$  must be odd  
 D.  $xyz$  must be even  
 E.  $x^2+y^2+z^2$  must be even

Label 40  
 40  $x$  is a positive integer such that  $x^2 + 2x + 7$  is even. What are the possible values of  $x$ ?

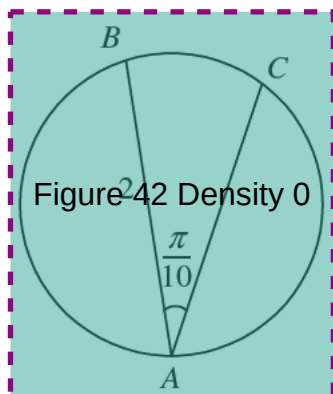
- A.  $x$  can be any positive integer  
 B.  $x$  can be any positive even number  
 C.  $x$  can be any positive odd number  
 D.  $x$  must be an even number greater than 10 000  
 E.  $x$  must be an positive odd number greater than 10 000

Label 37  
 37 If  $x^2 - kx + 9 \geq 0$  for all real values of  $x$ , what is the value of  $k$ ?

Label 41  
41 The perimeter of a sector is 16 and its angle is 2 radians. What is the area of the sector?

- A. 16
- B. 32
- C. 64
- D.  $16\pi$
- E.  $32\pi$

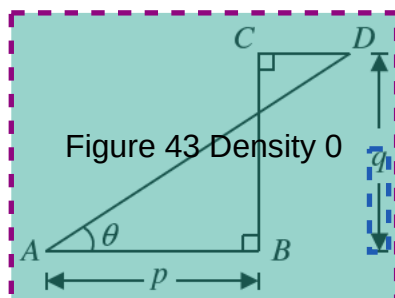
Label 42  
42



In the figure, diameter  $AB = 2$ ,  $\angle CAB = \frac{\pi}{10}$  rad. Minor arc  $BC =$

- A.  $\frac{\pi}{10}$
- B.  $\frac{\pi}{5}$
- C.  $\frac{3\pi}{10}$
- D.  $\frac{4\pi}{5}$
- E.  $\frac{9\pi}{10}$

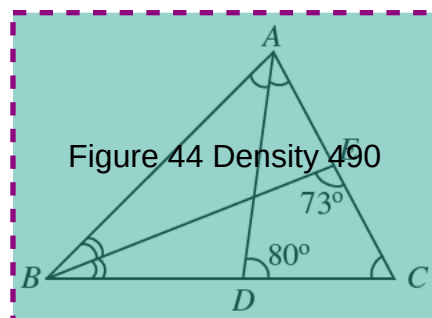
Label 43  
43



In the figure,  $\angle B = \angle C = 90^\circ$ . If  $AB = p$  and  $BC = q$ , then  $CD =$

- A.  $p + q \tan \theta$
- B.  $p + \frac{q}{\tan \theta}$
- C.  $p + q \cos \theta$
- D.  $p + q \tan \theta$
- E.  $p + \frac{q}{\tan \theta}$

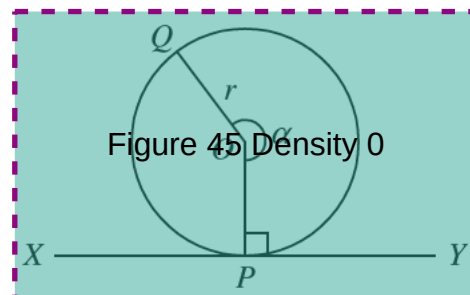
Label 44  
44



In the figure,  $AD$  and  $BE$  bisect  $\angle A$  and  $\angle B$  respectively.  $\angle C =$

- A.  $50^\circ$
- B.  $68^\circ$
- C.  $74^\circ$
- D.  $78^\circ$
- E.  $80^\circ$

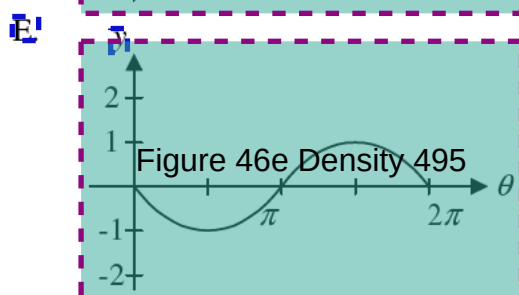
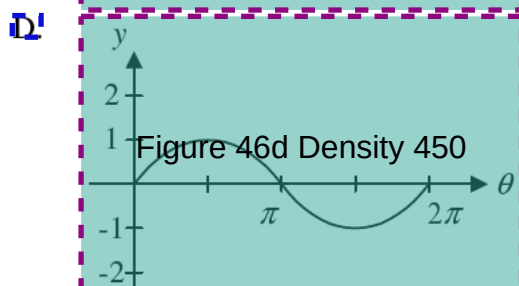
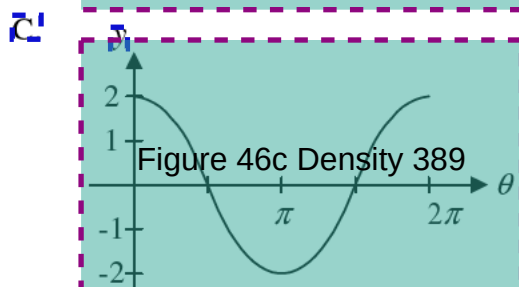
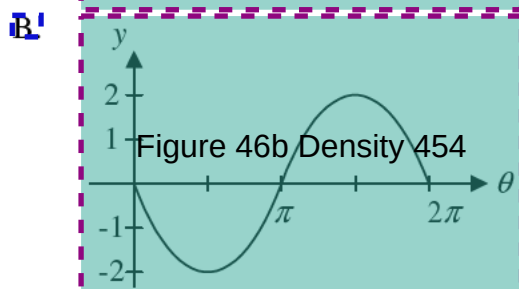
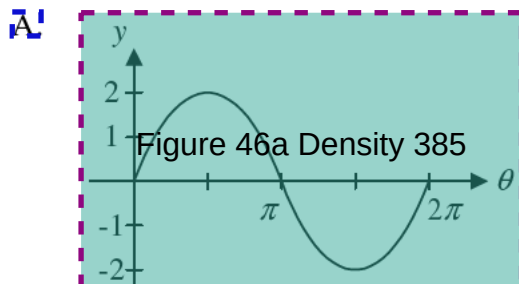
Label 45  
45



In the figure,  $O$  is the centre of the circle and its radius is  $r$ .  $XY$  touches the circle at  $P$ . Find the distance of  $Q$  from  $XY$ .

- A.  $r(1 - \sin \alpha)$
- B.  $r(1 + \sin \alpha)$
- C.  $r(1 - \cos \alpha)$
- D.  $r(1 + \cos \alpha)$
- E.  $r(2 - \sin \alpha)$

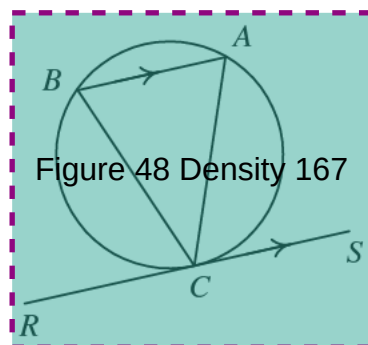
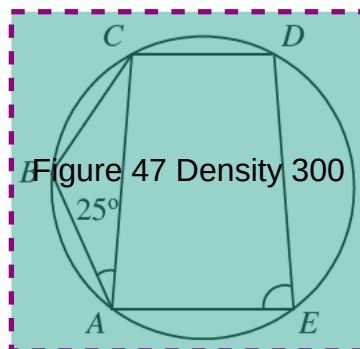
Label 46 Which of the following is the graph of  $y = 2 \sin \theta$ , where  $0 \leq \theta \leq 2\pi$ ?



Label 48

- A.  $50^\circ$   
 B.  $65^\circ$   
 C.  $75^\circ$   
 D.  $90^\circ$   
 E.  $105^\circ$

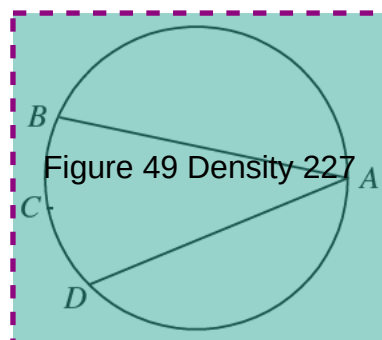
In the figure,  $AB = BC = CD$ .  $\angle AED =$



In the figure,  $RS$  is a tangent to the circle at  $C$ .  $BA$  is any chord parallel to  $RCS$ . Which of the chords  $AB$ ,  $BC$  and  $CA$  must be equal in length?

- A.  $AB$  and  $BC$  only  
 B.  $AC$  and  $BC$  only  
 C.  $AB$  and  $AC$  only  
 D. All of them  
 E. No two of them

Label 49





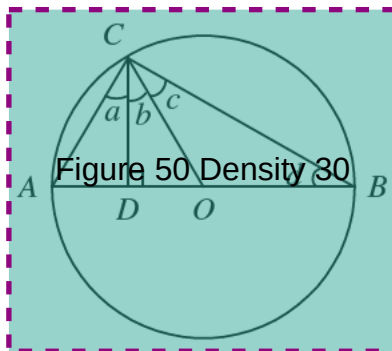
In the figure,  $AB = AC$ ,  $D$  is the mid-point of arc  $BC$ . Which of the following is/are true?

- I.  $AD$  bisects  $\angle BAC$
- II.  $BC \perp AD$
- III.  $AD$  is a diameter of the circle

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Figure 49 Density 4297

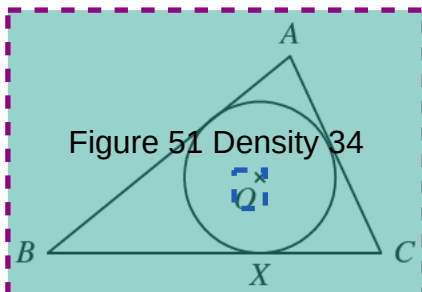
Label 50



In the figure,  $AOB$  is a diameter of the circle, centre  $O$ .  $CD$  is the perpendicular bisector of  $OA$ . Which of the angles  $a, b, c, d$  is/are equal to  $30^\circ$ ?

- A.  $a$  only
- B.  $a$  and  $b$  only
- C.  $a, b$  and  $c$  only
- D.  $a, b, c$  and  $d$
- E. None of them

Label 51

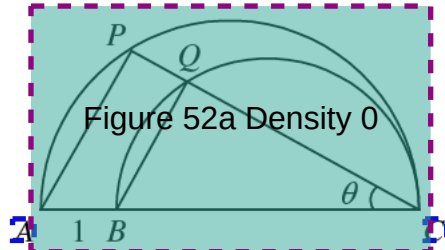


In the figure, circle  $O$  is inscribed in  $\triangle ABC$ , touching  $BC$  at  $X$ . Which of the following must be true?

- I.  $OX \perp BC$
- II.  $OA$  bisect  $\angle A$
- III.  $AO$  produced bisect  $BC$

- A. I only
- B. I and II only
- C. I and III only
- D. I, II and III only
- E. None of them

Label 52

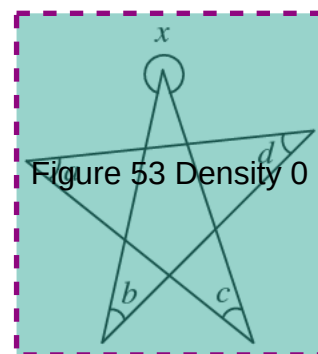


In the figure,  $AC$  and  $BC$  are diameters of two semi-circles touching each other internally at  $C$ .  $PQC$  is a straight line. If  $AB = 1$ , then  $PQ =$

- A.  $\cos \theta$
- B.  $\sin \theta$
- C.  $\tan \theta$
- D.  $\frac{1}{\sin \theta}$
- E.  $\frac{1}{\cos \theta}$

Figure 52b Density 3036

Label 53



With the notation in the figure, express  $a + b + c + d$  in terms of  $x$ .

- A.  $x - 180^\circ$
- B.  $x$
- C.  $540^\circ - x$
- D.  $360^\circ - x$

E.  $180^\circ - x$

Label 54  
54

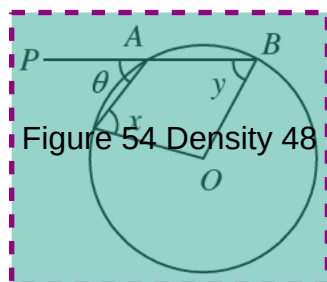


Figure 54 Density 48

In the figure,  $O$  is the centre of the circle.  $PAB$  is a straight line.  $x + y =$

- A.  $2\theta$
- B.  $90^\circ + \theta$
- C.  $180^\circ - \theta$
- D.  $180^\circ - 2\theta$
- E.  $180^\circ$